

ATPL EASA STUDY GUIDE | 040 - HUMAN PERFORMANCE

Chapter 5: Vision & Visual Illusions in Flight

Syllabus Source: CAE Oxford (pp. 109-158) | Pass Target: >= 90% AviationExam

1. FUNCTIONAL ANATOMY OF THE EYE

Vision provides approximately 80% to 90% of all sensory information used by a pilot in flight. Light enters through the cornea, passes through the pupil (regulated by the iris), is focused by the crystalline lens (controlled by ciliary muscles during accommodation), and projects an inverted image onto the retina.

Anatomical Structure	Primary Physiological Role	Aeronautical Significance
Cornea	Refracts light	Fixed optical element providing ~70% of total refractive power of the eye.
Crystalline Lens	Accommodation	Variable focus. Ciliary muscle contraction relaxes suspensory ligaments, making lens rounder for near vision.
Fovea Centralis	Sharp visual acuity	Small central pit in retina (macula lutea). Contains EXCLUSIVELY CONES (no rods). High-acuity photopic daytime vision.
Optic Disc	Blind Spot	Point where optic nerve exits retina and blood vessels enter. Lacks photoreceptors, creating an anatomical blind spot ~15° temporal.

2. PHOTORECEPTORS: RODS VS CONES (MASTER COMPARISON)

Characteristic	Cones (Daylight Vision)	Rods (Night Vision)
Quantity & Distribution	~6 to 7 million. Concentrated in Fovea Centralis.	~120 million. Absent in fovea; spread across peripheral retina.
Lighting Condition	Photopic Vision (Daylight, bright light).	Scotopic Vision (Night, low illumination). Mesopic = dusk/dawn.
Visual Acuity	Very HIGH acuity (sharp, detailed vision).	POOR visual acuity; highly sensitive to movement and dim light.
Color Vision	Trichromatic: Red, Green, Blue pigments.	Monochromatic: Black and white / shades of grey only.
Dark Adaptation Rate	Fast: Reaches maximum sensitivity in 5-7 min.	Slow: Rhodopsin (visual purple) requires 30 MINUTES to fully adapt!

[!] EXAM TRAP (AVIATIONEXAM) - The Night Blind Spot & Off-Center Viewing Technique

- The Night Blind Spot: At night, cones in the fovea centralis become inoperative due to lack of light. Because there are NO RODS in the fovea, the central 1° to 2° of the visual field is COMPLETELY BLIND at night!
- Off-Center Viewing: To detect dim objects or other aircraft at night, the pilot must NEVER look directly at them. Look 10° to 15° OFF-CENTER so that light falls onto the peripheral retina where rods are dense!

3. OPTICAL DEFECTS & EMPTY FIELD MYOPIA

- Myopia (Short-sightedness): Eyeball too long or lens too strong. Focal point falls in front of retina. Corrected with CONCAVE lenses (minus diopters).
- Hypermetropia (Long-sightedness): Eyeball too short. Focal point falls behind retina. Corrected with CONVEX lenses (plus diopters).
- Presbyopia: Aging-related hardening of crystalline lens, reducing accommodation for near vision. Corrected with reading glasses.
- Empty Field Myopia (Space Myopia): In featureless sky (overcast, haze, darkness), the eye automatically rests at a focal distance of 1 to 2 METERS! Distant aircraft will not be focused. Prevention: Regularly scan distant wingtips or ground features.

4. MASTER RUNWAY & APPROACH VISUAL ILLUSIONS

Visual Illusion Condition	Perceived Aircraft Position	Pilot Reaction & Flight Safety Hazard
Narrow Runway	Higher than actual	Pilot perceives aircraft is too high -> flies lower than normal approach -> risks undershooting / landing short.
Wide Runway	Lower than actual	Pilot perceives aircraft is too low -> flies higher approach -> flares too high, risks tailstrike / long landing.
Upsloping Runway	Higher than actual	Upslope gives visual illusion of being too high -> pilot pitches down, flies flatter approach, risks CFIT.
Downsloping Runway	Lower than actual	Downslope gives illusion of being too low -> pilot pitches up, flies steeper approach, lands long.
Black Hole Approach	Higher than actual	Night approach over water/dark terrain with no peripheral cues. Pilot flies dangerously low, curved approach into terrain.

[*] OFFICIAL DEFINITION - Sensory Illusions: Autokinesis & Flicker Vertigo

- Autokinesis: Staring continuously at a single static light in complete darkness causes the light to appear to move after several seconds due to micro-saccadic eye movements. Prevention: Keep eyes scanning; avoid fixating on a single point.
- Flicker Vertigo: Exposure to flashing light (sunlight through rotating propeller or strobe lights) at 4 to 20 Hz can induce nausea, vomiting, dizziness, loss of consciousness, and epileptic seizures.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): SHELL Model & Error Analysis

REGLA MNEMOTÉCNICA: L-H-S-E-L Paso 1: Identificar siempre el Liveware Central (L) como el piloto/operador humano. Paso 2: Si el problema involucra mandos, pantallas o botones confusos -> Es un desajuste L-H (Hardware). Paso 3: Si involucra una checklist mal redactada o un manual confuso -> Es un desajuste L-S (Software). Paso 4: Si involucra turbulencia, calor o hipoxia en cabina -> Es un desajuste L-E (Environment). Paso 5: Si involucra mala coordinación entre Capitán y Primer Oficial o con ATC -> Es un desajuste L-L (Liveware).

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: How long does it take for rod photoreceptors in the human retina to achieve full dark adaptation?

- [A]: 5 to 7 minutes
- [B]: 10 to 15 minutes
- [C]: Approximately 30 minutes
- [D]: 2 hours

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. Cones adapt quickly in 5 to 7 minutes, but rod rhodopsin regeneration requires approximately 30 minutes for complete dark adaptation.

Q2: What scanning technique should a pilot employ to detect traffic visually at night?

- [A]: Stare fixedly at the central area where the object is suspected
- [B]: Look 10° to 15° off-center and scan peripherally
- [C]: Use bright white flashlight inside the cockpit
- [D]: Rapidly blink both eyes continuously

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. The fovea centralis has no rods and is blind at night (night blind spot). Looking 10° to 15° off-center projects light onto rod-rich peripheral retina.

Q3: When approaching a runway that is significantly narrower than normal, what visual illusion is created?

- [A]: Illusion of being lower than actual, causing the pilot to fly a high approach
- [B]: Illusion of being higher than actual, causing the pilot to fly a lower approach and risk undershooting
- [C]: Illusion of excessive airspeed
- [D]: Illusion of bank angle

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. A narrow runway subtends a smaller visual angle, leading the pilot to believe they are higher than actual, causing a flatter approach and undershoot risk.

Q4: What is 'Empty Field Myopia' and what is the typical resting focus distance of the eye in a featureless sky?

- [A]: Loss of peripheral vision above FL 300
- [B]: Accommodation resting at a distance of 1 to 2 meters in the absence of visual focal targets
- [C]: Permanent damage caused by UV radiation
- [D]: Inability to read cockpit instruments in bright sunlight

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Without focal stimuli, ciliary muscles relax to a resting accommodation distance of approximately 1 to 2 meters (3 to 6 feet).

Q5: What optical lens is prescribed to correct myopia (short-sightedness)?

- [A]: Convex lenses (plus diopters)
- [B]: Concave lenses (minus diopters)
- [C]: Cylindrical prisms only
- [D]: Polarized dark filters

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Myopia focuses light in front of the retina; concave (diverging) lenses move the focal point back onto the retina.

Q6: How does smoking cigarettes affect a pilot's visual acuity at night?

- [A]: It improves dark adaptation due to nicotine stimulation
- [B]: It has no measurable effect below 10,000 ft
- [C]: Carbon monoxide binds to hemoglobin, raising physiological altitude by 4,000-5,000 ft and severely degrading night vision
- [D]: It causes immediate temporary color blindness

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. Smoking creates 4-8% carboxyhemoglobin, inducing hypoxia in oxygen-hungry retinal rods and significantly degrading night vision even at sea level.

Q7: What visual illusion does heavy rain on the cockpit windshield produce during final approach?

- [A]: Creates an illusion of being higher than actual, causing the pilot to pitch down and fly dangerously low
- [B]: Creates an illusion of being lower than actual, causing the pilot to fly higher
- [C]: Has no effect on perceived glideslope
- [D]: Flattens the runway perspective completely

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [A]. Water refraction bends light rays downward, making the horizon appear lower and creating the illusion that the aircraft is higher than actual. The pilot tends to fly a flatter, dangerously low approach.

Q8: Which ocular structure actively changes shape to focus near and far objects during visual accommodation?

- [A]: The cornea
- [B]: The crystalline lens, modulated by the ciliary muscle
- [C]: The vitreous humor
- [D]: The optic disc

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Contraction and relaxation of the ciliary muscle adjusts the curvature of the crystalline lens to achieve visual focus on near and distant targets.